

IoT BidWars 2026 – Summary Report

IoT Applications Club (IoTA) & Department of Instrumentation &
Electronics Engineering

Jadavpur University, Kolkata, India

*In Collaboration with IEEE Circuits & Systems Society (IEEE CAS)
& International Society of Automation (ISA)*

at

Srijan 2026, Jadavpur University

Abstract — IoT BidWars 2026 is a multi-round, strategy-intensive technical competition organized by the IoT Applications Club (IoTA) and the Department of Instrumentation & Electronics Engineering, Jadavpur University, in collaboration with IEEE Circuits & Systems Society (CAS) and the International Society of Automation (ISA), as part of Srijan 2026 — the Annual Techno-Management Fest of Jadavpur University presented by F.E.T.S.U. Themed “Strategic Warfare Meets Smart Circuitry,” the event challenged participants to combine strategic resource management with applied digital electronics and IoT engineering. The competition spanned two days and three rounds: an online preliminary quiz, an offline digital component auction and circuit-build challenge, and an offline IoT-based cloud connectivity task. A total of 75 teams comprising approximately 187 participants cleared the preliminary stage, with 21 teams qualifying for the offline rounds and 8 teams competing in the grand finale. IoT BidWars fostered interdisciplinary collaboration, sharpened competitive hardware skills, and drew participation from over 25 institutions across India.

I. INTRODUCTION

The Internet of Things (IoT) has emerged as a pivotal enabling technology for developing intelligent systems that integrate sensing, actuation, communication, and data-driven decision-making across domains such as healthcare, transportation, agriculture, smart infrastructure, and industrial automation. Alongside IoT, foundational digital electronics knowledge — encompassing logic design, circuit construction, and hardware interfacing — remains a bedrock engineering skill. Cultivating these competencies through competitive, hands-on environments is essential for producing industry-ready graduates.

The IoT Applications Club (IoTA) at Jadavpur University is the official student-run technical society dedicated to exploring and advancing the IoT and embedded systems ecosystem. Founded in 2023, IoTA operates as a collective of visionary engineers united by a shared mission: to bridge the gap between theoretical knowledge and practical hardware innovation. The club is supported by the IEEE Circuits & Systems Society (IEEE CAS), Jadavpur University Chapter, and functions under the mentorship of senior faculty from the Department of Electronics & Telecommunication Engineering, Jadavpur University.

IoTA's activities span three core pillars: (1) Advanced R&D — developing low-latency sensor networks and embedded smart systems; (2) Knowledge Fusion — conducting deep-dive workshops on Embedded C, PCB design, and RTOS to make members industry-ready; and (3) Robotics & AI — merging mechanical precision with intelligent algorithms to build autonomous, perception-driven systems.

Since its inception, IoTA has organized a range of technical events including Connexion 2025 (an IoT ideation competition), Robotic Workshops, PCB Design Bootcamps, and the flagship Robo-Sumo competition Lord of the Ring, now in its third edition. The club's vision is to become the epicentre of IoT research and product development at Jadavpur University and to foster a culture of radical technical excellence and professional growth among students.

II. ABOUT SRIJAN 2026 & IOT BIDWARS

Srijan is the Annual Techno-Management Fest of Jadavpur University, organized by the Faculty of Engineering and Technology Students' Union (F.E.T.S.U.). One of the premier collegiate technical festivals in Eastern India, Srijan has grown from a modest collection of student-run events since its founding in 2007 into a national-scale festival hosting over 50 events annually — spanning coding, robotics, gaming, management, and innovation. A defining hallmark of Srijan is that participation is entirely free of charge, setting it apart from most major national techfests and ensuring maximum student accessibility.

IoT Bidwars 2026 is a competition conceptualized and organized by IoTA in partnership with the Department of Instrumentation & Electronics Engineering (IEE), Jadavpur University. The event was designed to test participants not only on their technical knowledge and hardware skills but also on their strategic thinking and resource management. Participants were challenged to outbid rivals for limited electronic components in a live auction, construct functional digital logic circuits from scratch, and interface sensors with microcontrollers to stream live data to the cloud — all under real-time pressure and competition. Registrations were accepted through the IoTA club website and the Unstop platform.

III. EVENT STRUCTURE

IoT BidWars 2026 was structured as a two-day, three-round competition designed to progressively evaluate participants across theoretical knowledge, circuit-building strategy, and IoT implementation depth.

A. Competition Format

- Multi-round competition spanning two days, combining online and offline modes.
- Round 1 (Online): Preliminary Quiz open to all registered participants.
- Round 2A (Offline): Bid-n-Build — component auction followed by digital circuit construction.
- Round 2B (Offline): Uplink — IoT sensor interfacing and real-time cloud data transmission.
- Team size: 2–3 members per team.
- All decisions of the judging panel are final and binding.

B. Rounds

1. Round 1 — Preliminary Quiz (Online)

- Open registration via the IoTA club website and the Unstop platform.
- Timed MCQ quiz covering Boolean Logic, Mathematical Reasoning, and Fundamentals of Electronics.
- Shortlisting based on scores; ties resolved by time of submission.

2. Round 2A — Bid-n-Build (Offline)

- Only teams shortlisted from Round 1 qualified for this stage.
- Before the auction, each team received three problem statements to review and plan for.
- Each team was allotted a fixed virtual budget of ₹10,000.
- Digital electronics components — Logic ICs including the 74153 (4-to-1 Multiplexer), 7408 (Quad AND Gate), 7404 (Hex Inverter), and 7432 (Quad OR Gate) — were made available through a live multi-round auction.
- Teams bid strategically round by round; inter-team component trading was also permitted under defined rules.
- Using exclusively the components they acquired, each team selected one problem statement and constructed the required digital circuit.
- Evaluation criteria: circuit accuracy, component-use efficiency, completion time, and overall implementation quality.

3. Round 2B — Uplink (Offline)

- Conducted and evaluated entirely by the IoT Applications Club.
- Each team received a standardised hardware kit comprising an ESP32 microcontroller, PIR (Passive Infrared) motion sensor, LED, buzzer, LCD display, voltage regulator, resistors, breadboard, connecting wires, and a battery.
- Objective: detect motion using the PIR sensor via the ESP32, drive an LED and buzzer as output, display readings on the LCD, and transmit real-time sensor data to the cloud.
- Teams were encouraged to demonstrate innovation: one team displayed motion count and estimated distance on the LCD as a novel enhancement; another used a Blynk graph widget for live cloud data visualisation.
- Out of 8 finalist teams: 5 successfully acquired and interfaced the PIR sensor; of those, 4 completed the LED/buzzer output stage; and 3 achieved full integration including live cloud data transmission.
- Evaluation criteria: working hardware setup, successful cloud data transmission, reading accuracy, system stability, and execution time.

C. Evaluation Criteria

- Innovation and originality of approach
- Technical accuracy and functional correctness
- Strategic and efficient use of acquired components and budget
- System design quality and completeness
- Cloud integration and data transmission quality (Uplink round)
- Clarity, stability, and speed of execution

D. Final Scoring

Final Score = 50% (Bid-n-Build) + 50% (Uplink). The team with the highest combined score was declared the Overall Winner. In the event of a tie, total execution time served as the deciding tiebreaker.

IV. EVENT TIMELINE

A. Round 1 — Preliminary Quiz

Duration	Mode	Registered Teams	Details
9th April 2026 (Saturday)	Online	~116 (240+ participants)	Open MCQ quiz via Unstop & IoTA website; ~75 teams attempted; 21 shortlisted

B. Round 2 — Finals (Bid-n-Build + Uplink)

Date	Mode	Finalist Teams	Details
10th April 2026 (Friday)	Offline — JU Campus	8 Teams	Live component auction, digital circuit build, and IoT cloud challenge; final scores computed and prizes awarded

V. PARTICIPATION STATISTICS

Stage	Teams	Participants
Total Registrations (Website + Unstop)	~116	~240+
Prelims Attempted (Round 1)	75	~187
Shortlisted for Offline Rounds	21	~48
Grand Finalists	8	~20

Participation Funnel - IoT BidWars 2026

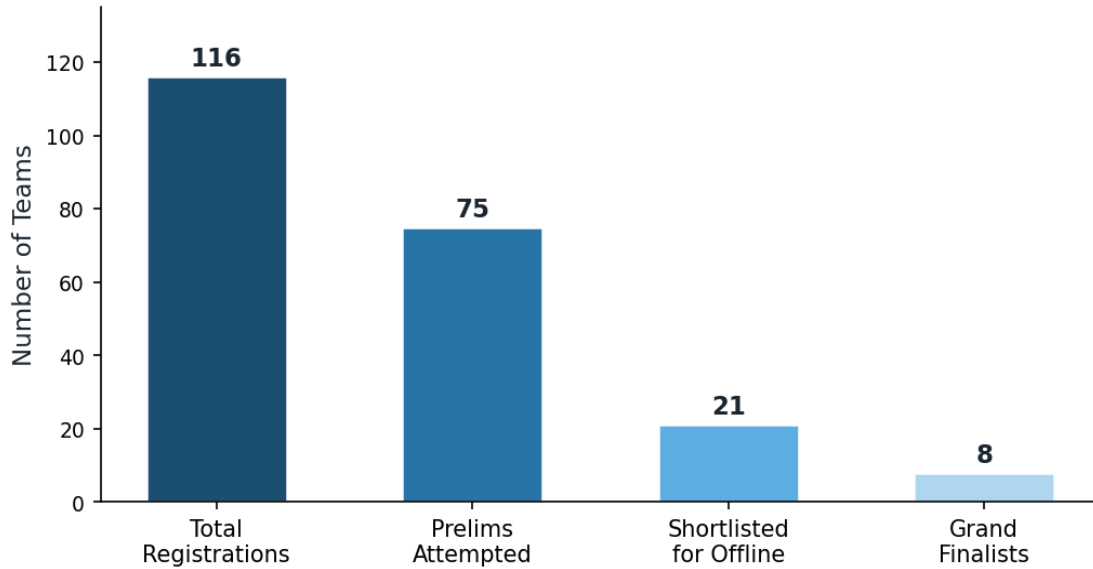


Figure 1: Participation Funnel — IoT BidWars 2026

VI. GEOGRAPHIC DIVERSITY & FOOTFALL

Participants registered from over 25 institutions across India, demonstrating the significant outreach and cross-institutional appeal of IoT BidWars 2026. The table below captures the top institutions by participant count from the preliminary round data.

#	Institution	Location	Participants (Approx.)
1	Jadavpur University	Kolkata, West Bengal	~90
2	Academy of Technology (AoT)	Hooghly, West Bengal	~19
3	Heritage Institute of Technology (HITK)	Kolkata, West Bengal	~15
4	Hooghly Engineering & Technology College (HETC)	Hooghly, West Bengal	~15
5	University of Calcutta	Kolkata, West Bengal	~13
6	Dr. B.C. Roy Engineering College (BCREC)	Durgapur, West Bengal	~11
7	Narula Institute of Technology	Kolkata, West Bengal	~6

#	Institution	Location	Participants (Approx.)
8	Kalyani Government Engineering College (KGEC)	Nadia, West Bengal	~6
9	IIST Shibpur	Howrah, West Bengal	~5
10	Techno Main Salt Lake	Kolkata, West Bengal	~3
11	IIT Bhilai	Chhattisgarh (Outside WB)	~3
12	Sant Longowal Institute (SLIET)	Punjab (Outside WB)	~1
13	ICT Mumbai / LPU / Others	Outside West Bengal	~7

Of the total registered participants, approximately 47% were from Jadavpur University, 44% from other West Bengal institutions, and 9% from institutions outside West Bengal — spanning states including Chhattisgarh, Punjab, Maharashtra, Tamil Nadu, Uttarakhand, and Jharkhand. This geographic diversity reflects the growing national visibility of IoTA's events.

Geographic Distribution of Participants (Prelims - IoT BidWars 2026)

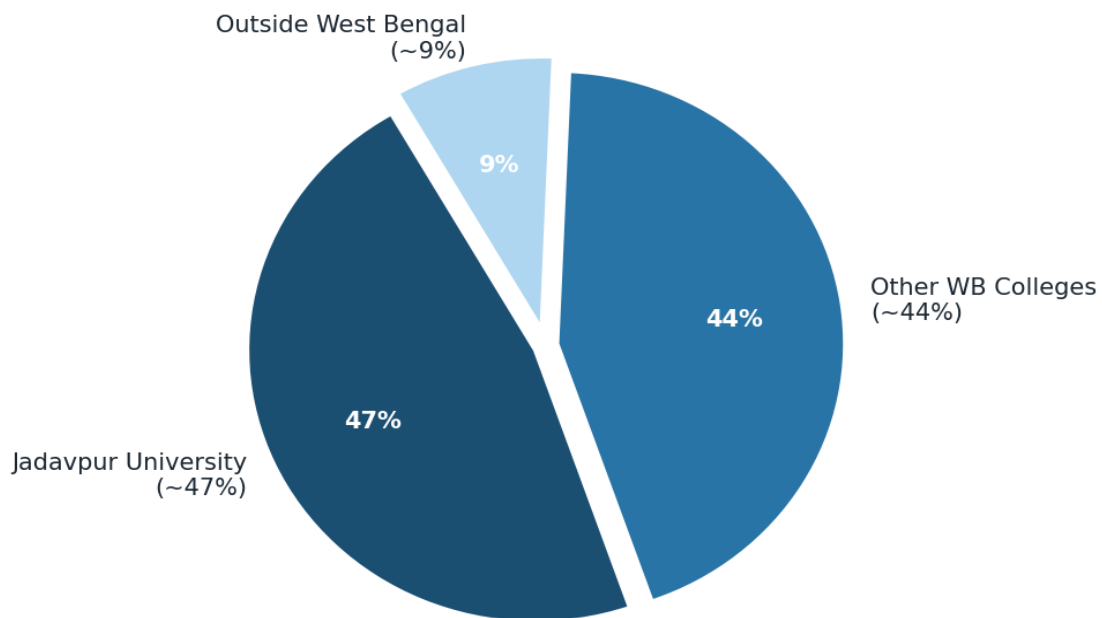


Figure 2: Geographic Distribution of Participants — IoT BidWars 2026 (Prelims)

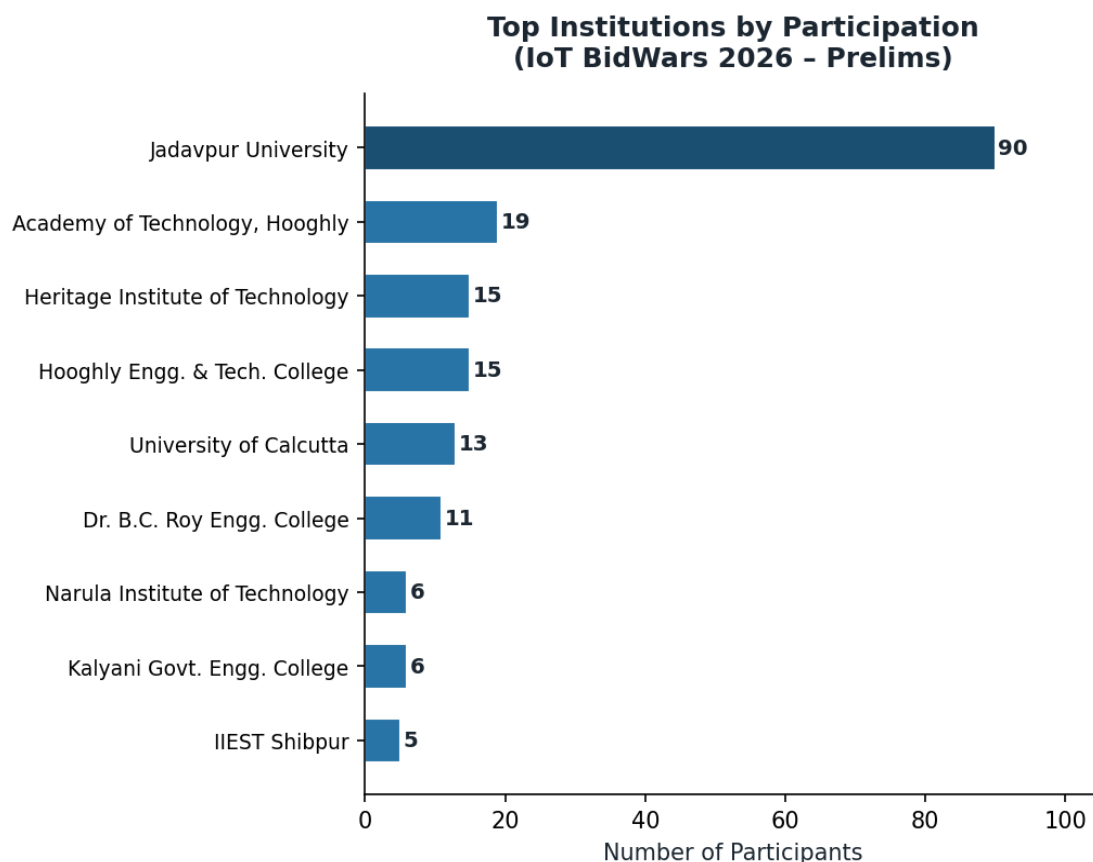


Figure 3: Top Institutions by Participation — IoT BidWars 2026 (Prelims)

VII. QUALIFIER TEAMS

#	Team Name	Members	Institution
1	Agrinomics 2.0	Apurba Mandal, Shreyam Marodia, Arpan Karmakar	Jadavpur University
2	Antarez	Aayushman Das, Rahul Ganguly	Jadavpur University / University of Calcutta
3	Binary Barons	Soham Majhi, Sabyasachi Majumdar, Sidhorishi Das	Jadavpur University
4	Cache Crushers	Ranit Nandi, Soham Mukherjee, Soumik Das	Techno Main Salt Lake
5	Circuit Bidders	Sourasish Mitra, MD Ceezan Ahmed, Aghnya Kundu	Jadavpur University
6	Circuit Breakers ⚡	Sourya Sirsha Mukhopadhyay, Soumajit Saha	IEM / Heritage IT

#	Team Name	Members	Institution
7	Default (TechTonic)	Arghadip Dutta, Advay Mondal, Bidisha Sinha	JU / Heritage IT
8	Digital Dynamos	Rajorshi Ghosh, Indranil Laha, Debanjan Goswami	Hooghly Engg. & Tech. College
9	ElectraX	Mainak Bera, Sandipan Maity, Nabhendu Bera	Jadavpur University
10	Error 404: Not Found	Avik Rana, Syamantak Mondal, Sayandip Bhuin	Jadavpur University
11	evolvEX	Amit Gupta, Debdatta Kundu, Rishav Shaw	Jadavpur University
12	IoTians	Subhajit Ghosh, Survi Mukherjee, Rimi Sarkar	Dr. B.C. Roy Engg. College, Durgapur
13	IoTix	Rahul Chatterjee, Chirantan Ghosh, Pritha Bose	Hooghly Engg. & Tech. College
14	Logic Loops	Sumit Pal, Supriyo Paramanik, Manabendra Mondal	Narula Institute of Technology
15	MR360	Rahul Hossain, Mosaddir Momin, Suparna Nanda G.	Jadavpur University
16	Neural Nexus	Ayan Chandra, Bitan Saha	Jadavpur University
17	RoboWarriors 1.0	Biswajit Karan, Soumajit Boral, Rohit Kumar Roy	Jadavpur University
18	RSC Legends	Srayan Dutta, Anjishnu Sinha, Soham Kar	University of Calcutta
19	Tech Idiots	Aurthit Paul, Rishav Pramanick	University of Calcutta
20	Triode	Krishanu Jyoti Saradar, Saswata Das, Malay Manna	Jadavpur University
21	12 SAMOSE	Sourav Mahato, Alok Kumar Roy	Dr. B.C. Roy Engg. College

VIII. JUDGING PANEL

Judges:

Mr. Indranil Das Proprietor & CEO, Synchro Electronics, Kolkata *Synchro Electronics is an ISO 9001:2015 certified Automatic Test Equipment (ATE) manufacturer established in 1993. Mr. Das holds a PhD from IIST Shibpur and is associated with the Centre of Excellence for Green Energy & Sensor Systems, IIST Shibpur. The company has contributed specialised products in instrumentation, automation, and calibration to industries across eastern India.*

Mr. Sandip Sarkar Founder & CIO, XCONICS, Kolkata *XCONICS is a product-based company providing 360° solutions to startups and MSMEs in electronics, IoT, and embedded systems development under a revenue-share model. The company maintains an Intellectual Property (IP) Bank and has conducted IoT training workshops at Jadavpur University in collaboration with Tata Steel and the West Bengal Academy of Science & Technology.*

Dr. Chandrima Mondal Assistant Professor, Department of Electronics & Telecommunication Engineering, Jadavpur University *Dr. Mondal's research focuses on the modelling and simulation of nanoelectronic devices and circuits, including work on TMD-based Negative Capacitance FETs. She is a recipient of the 2025 IEEE Region 10 Outstanding Volunteer Award and is associated with the IC Centre of Excellence at Jadavpur University.*

IX. WINNERS

Position	Team Name	Members	Institution
1st Place ₹7,000	QBIT DYNAMICS	Saaraswata Roy, Sarthak Sharan, Rohan Sarkar	Heritage Institute of Technology, Kolkata
2nd Place ₹5,000	TECH TITANS	Disha Das, Srijita Misra, Md Ayman Siddique	Kalyani Govt. Engineering College
3rd Place ₹3,000	CONNECTING THE WORLD	Debanjan Chakraborty, Ritarup Tripathy, Sohail Hasan	Jadavpur University

Position	Prize
1st Place	₹ 7,000
2nd Place	₹ 5,000
3rd Place	₹ 3,000
Total Prize Pool	₹ 15,000



Certificates of participation were awarded to all 8 finalist teams. All participants who cleared the preliminary round received digital certificates of achievement.

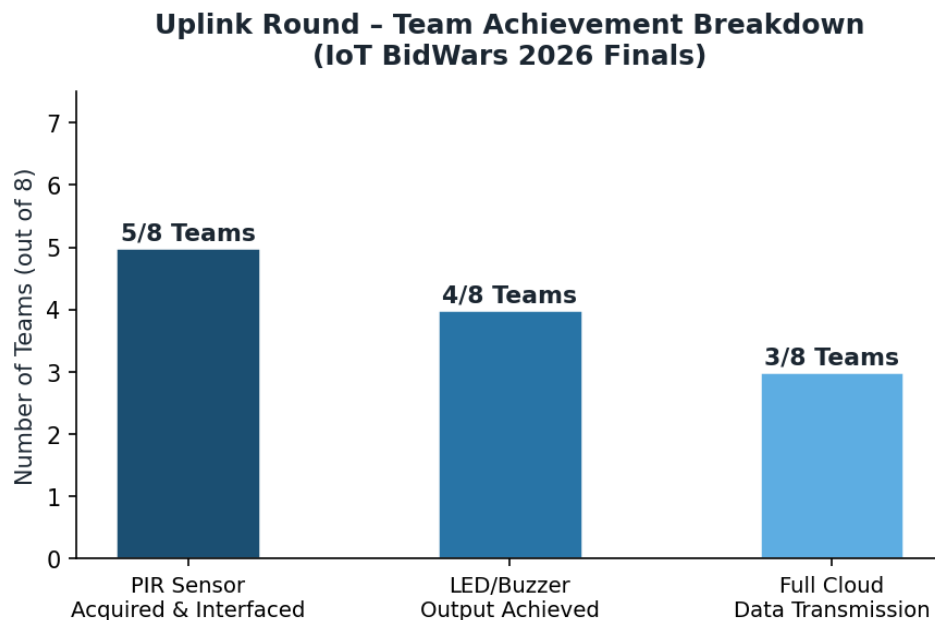


Figure 4: Uplink Round — Team Achievement Breakdown (IoT BidWars 2026 Finals)

X. FUTURE ASPECTS

- All finalist and participating teams were encouraged to continue developing their hardware and IoT projects, with an open invitation to reach out to IoTA for mentorship and continued support at any time.
- Participants were invited to engage in upcoming IoTA workshops, inter-club projects, and technical events planned throughout the academic year.
- IoT BidWars is envisioned to become a flagship annual event of the IoT Applications Club at Srijan, growing in scale, problem complexity, and participation with each successive edition.
- IoTA plans to deepen its collaboration with the International Society of Automation (ISA), the IEEE CAS Chapter, and the Department of IEE to offer more structured training in digital electronics, embedded systems, and cloud IoT.
- The club continues to advance its broader mission of producing a monthly mini-magazine covering recent developments in IoT and related fields, featuring research insights from the Department of Electronics & Telecommunication Engineering, Jadavpur University.
- Teams have been constituted within IoTA to represent the club at national and international hackathons and technical competitions in the coming year.

XI. ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to the University administration of Jadavpur University for providing us the opportunity to organise and host IoT BidWars 2026 as part of Srijan ...26. This event provided a vital platform for applied hardware engineering, strategic thinking, and IoT innovation among students from across India.

We extend our deepest thanks to our distinguished panel of judges for their time, expertise, and constructive evaluation of the finalist teams:

- **Mr. Indranil Das**, Proprietor & CEO, Synchro Electronics
- **Mr. Sandip Sarkar**, Founder & CIO, XCONICS
- **Dr. Chandrima Mondal**, Professor, Dept. of ETCE, Jadavpur University

We extend our heartiest gratitude to the faculty mentors of the IoT Applications Club, whose guidance and unwavering support have been the backbone of all IoTA initiatives:

- **Prof. Sheli Sinha Chaudhuri** — Professor, Dept. of Electronics & Telecommunication, Jadavpur University
- **Prof. Palaniandavar Venkateshwaran** — Professor, Dept. of Electronics & Telecommunication, Jadavpur University
- **Dr. Chinmoy Ghorai** — Assistant Professor, Dept. of Electronics & Telecommunication, Jadavpur University
- **Dr. Chandrima Mondal**, Professor, Dept. of ETCE, Jadavpur University.

We acknowledge the Department of Instrumentation & Electronics Engineering, Jadavpur University for co-organising this event, and the International Society of Automation (ISA) for their collaboration and support. We also thank F.E.T.S.U. and the Srijan ...26 organising committee for providing the platform that gave this event its scale and national reach.

We acknowledge the hard work of our Event Points of Contact — Aryan Singh (IEE), Tuhin Roy (ETCE), Md. Tajuddin (IEE), Soumojeet (ETCE), Soureen Majumdar (IEE), and Arnab Karmakar (ETCE) — along with all IoTA club members and volunteers whose dedication ensured the smooth and successful execution of the competition.

Finally, we extend our appreciation to every team that participated in IoT BidWars 2026 — from the 75+ teams that took on the preliminary quiz to the 8 finalists whose technical excellence, creative thinking, and competitive spirit made this inaugural edition a truly memorable and inspiring event.

Gallery



Pic 1 : The Judge Panel

Prof. Chandrima Mondal, Dept. of ETCE, Jadavpur University & Chair, IEEE CAS JU Student Chapter (Left)
 Mr. Sandip Sarkar, Founder & CIO, Xconics Innovations Pvt. Ltd (Centre)
 Mr. Indranil Das, Founder & CEO, Synchro Electronics (Right)



Pic 2 : Participant – Judge Interaction & Evaluation



Pic 3 : Participants preparing the circuits & cloud architecture. CAS logo visible on the flex (Right)



Pic 4 : Prize Distribution Ceremony



Pic 5 : Handing over Prizemoney to the Champions (Team Qbit Dynamics)



Pic 6 : IoT Bidwars Official Poster